实验报告

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<tr><td>题目</td><td colspan="3">实验二　DBMS 软件安装及数据定义、修改、导入</td></tr>
<tr><td>姓名</td><td>刘师妤</td><td>学号</td><td>2024201568</td></tr>
<tr><td colspan="4">实验环境（计算机配置，操作系统，编程语言，编程工具等）

硬件平台：MacBook Air（M5，24G）

操作系统：macOS

DBMS：PostgreSQL 17.11（Homebrew），同时进行了 MySQL 和 MariaDB 的安装与使用测试

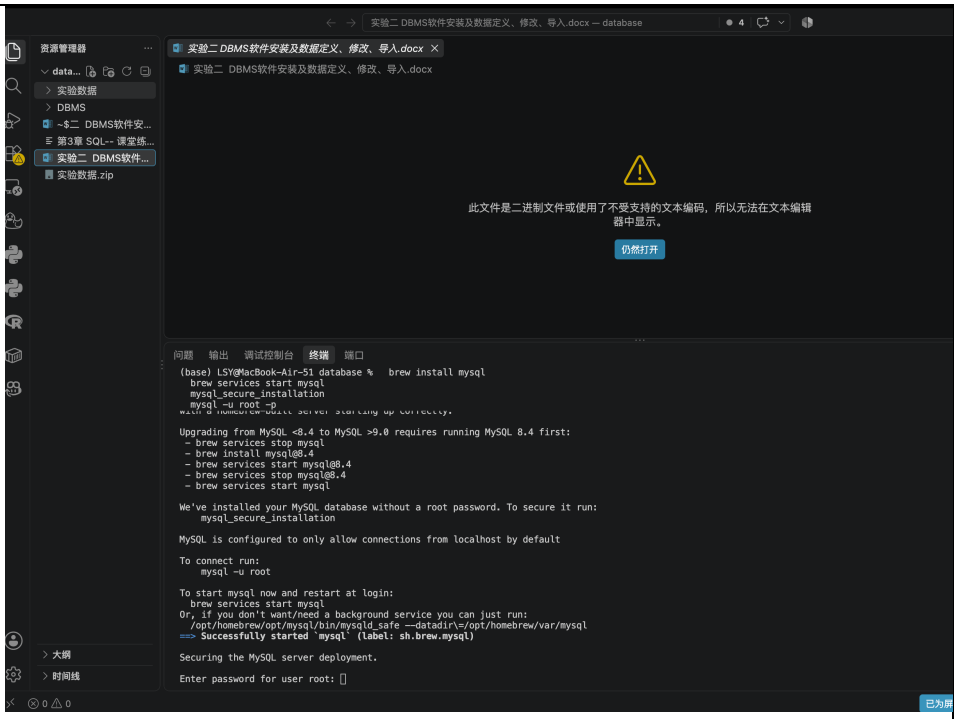
数据库：tpch

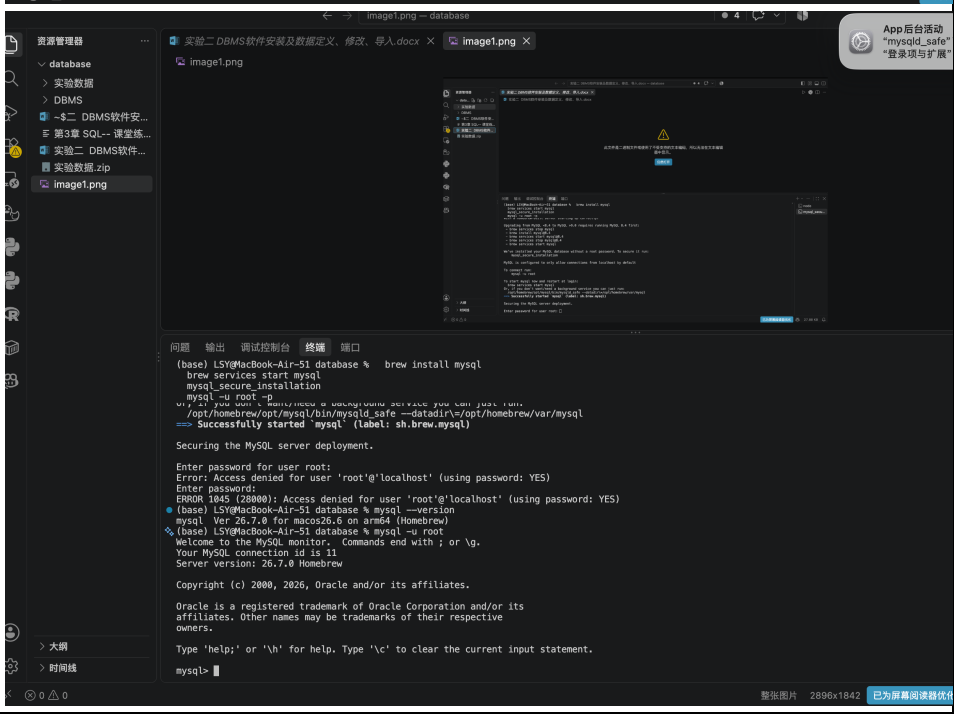
数据库语言：SQL

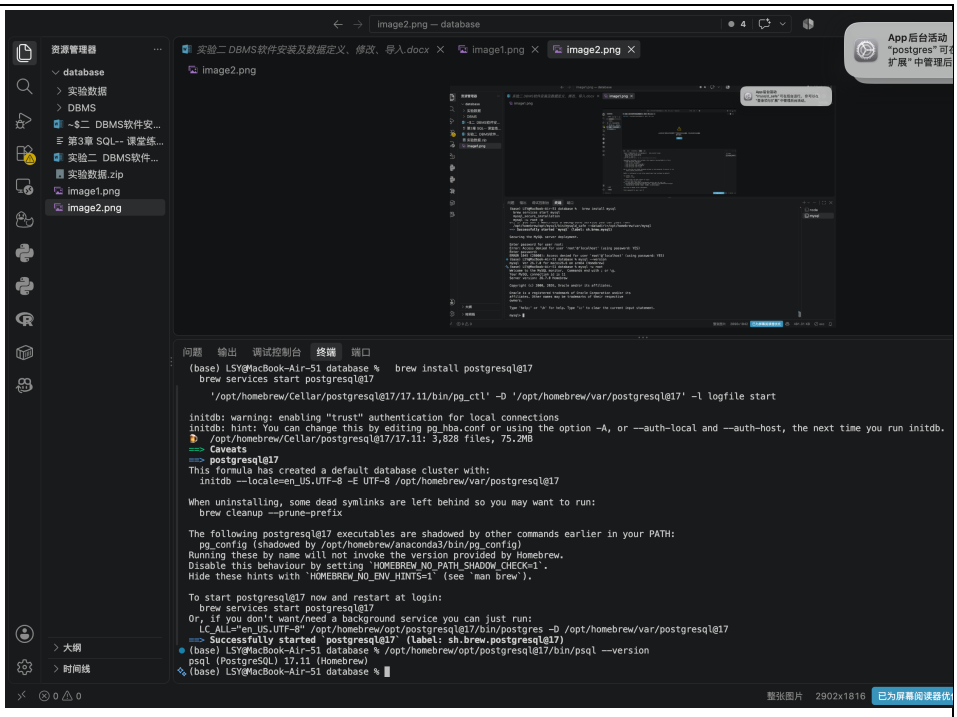
实验工具：Homebrew、PostgreSQL psql、VS Code/Terminal

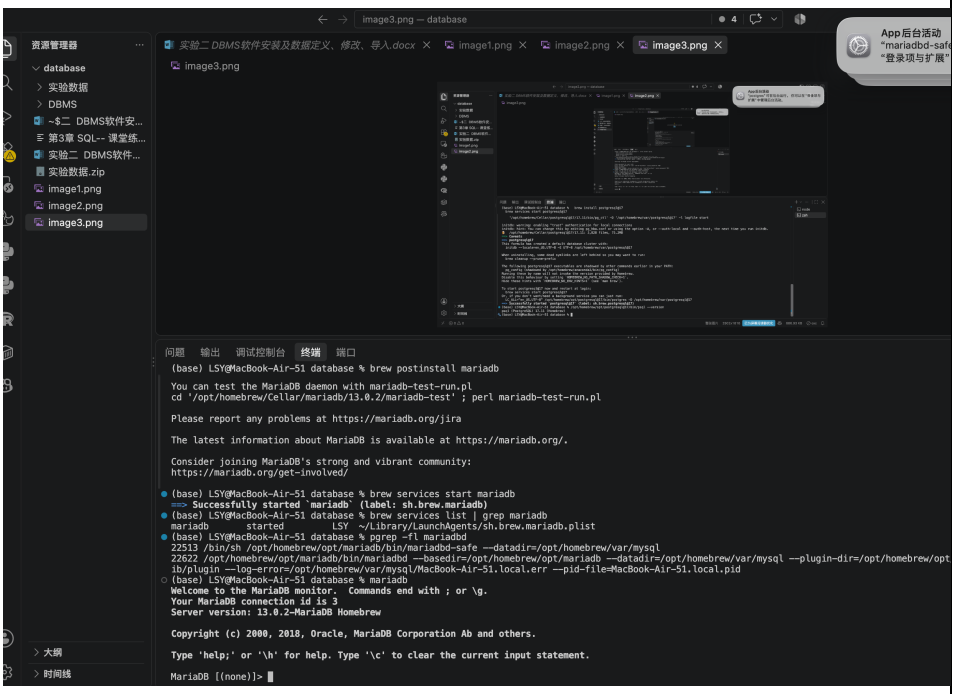
实验数据：TPC-H，Scale Factor = 0.01，采用实验提供的 CSV 数据文件</td></tr>
<tr><td colspan="4">实验结果及分析

1. 至少三个安装 DBMS 过程截图。</td></tr>
</table>


实验二 DBMS软件安装及数据定义、修改、导入.docx — database
资源管理器
实验二 DBMS软件安装及数据定义、修改、导入.docx
此文件是二进制文件或使用了不受支持的文本编码，所以无法在文本编辑器中显示。
仍然打开
问题 输出 调试控制台 终端 端口
(base) LSY@MacBook-Air-51 database % brew install mysql
brew services start mysql
mysql_secure_installation
mysql -u root -p
Upgrading from MySQL <8.4 to MySQL >9.0 requires running MySQL 8.4 first:
brew services stop mysql
brew install mysql@8.4
brew services start mysql@8.4
brew services stop mysql@8.4
brew services start mysql
We've installed your MySQL database without a root password. To secure it run:
mysql_secure_installation
MySQL is configured to only allow connections from localhost by default
To connect run:
mysql -u root
To start mysql now and restart at login:
brew services start mysql
Or, if you don't want/need a background service you can just run:
/opt/homebrew/opt/mysql/bin/mysqld_safe --datadir\=/opt/homebrew/var/mysql
==> Successfully started `mysql` (label: sh.brew.mysql)
Securing the MySQL server deployment.
Enter password for user root:
大纲
时间线


image1.png — database
资源管理器
image1.png
App后台活动
"mysqld_safe"
"登录项与扩展"
问题 输出 调试控制台 终端 端口
(base) LSY@MacBook-Air-51 database % brew install mysql
brew services start mysql
mysql_secure_installation
mysql -u root -p
/opt/homebrew/opt/mysql/bin/mysqld_safe --datadir\=/opt/homebrew/var/mysql
==> Successfully started `mysql` (label: sh.brew.mysql)
Securing the MySQL server deployment.
Enter password for user root:
Error: Access denied for user 'root'@'localhost' (using password: YES)
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
(base) LSY@MacBook-Air-51 database % mysql --version
mysql Ver 26.7.0 for macos26.6 on arm64 (Homebrew)
(base) LSY@MacBook-Air-51 database % mysql -u root
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 26.7.0 Homebrew
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>
大纲
时间线
整张图片 2896x1842





2. 执行 DDL、DML 语句的运行截图。

```
tpch=#   CREATE TEMP TABLE ddl_dml_demo (
      id INTEGER PRIMARY KEY,
      name VARCHAR(30)
  );

  INSERT INTO ddl_dml_demo VALUES (1, 'TPC-H');
  UPDATE ddl_dml_demo SET name = 'PostgreSQL TPC-H' WHERE id =
  SELECT * FROM ddl_dml_demo;
  DELETE FROM ddl_dml_demo WHERE id = 1;
CREATE TABLE
INSERT 0 1
UPDATE 1
 id |       name
----+------------------
  1 | PostgreSQL TPC-H
(1 行记录)

DELETE 1
tpch=#
```

```
取消发送的请求
tpch=# \dt
                关联列表
 架构模式 |   名称   |  类型  | 拥有者
----------+----------+--------+--------
 public   | customer | 数据表 | LSY
 public   | lineitem | 数据表 | LSY
 public   | nation   | 数据表 | LSY
 public   | orders   | 数据表 | LSY
 public   | part     | 数据表 | LSY
 public   | partsupp | 数据表 | LSY
 public   | region   | 数据表 | LSY
 public   | supplier | 数据表 | LSY
(8 行记录)
```

```
问题   输出   调试控制台   终端   端口

(base) LSY@MacBook-Air-51 database % psql tpch
      栏位       |          类型          | 校对规则 |  可空的  |
-----------------+------------------------+----------+----------+
 l_orderkey      | integer                |          | not null |
 l_partkey       | integer                |          | not null |
 l_suppkey       | integer                |          | not null |
 l_linenumber    | integer                |          | not null |
 l_quantity      | numeric(15,2)          |          | not null |
 l_extendedprice | numeric(15,2)          |          |
```

3. 数据生成导入的运行截图

```
DELETE 1
tpch=#   CREATE TEMP TABLE region_import_demo
  (LIKE region INCLUDING ALL);

  \copy region_import_demo FROM '/Applications/database/实验数据/csv/region.csv' WITH (F

  SELECT COUNT(*) FROM region_import_demo;
CREATE TABLE
COPY 5
 count
-------
     5
(1 行记录)
```

4. 在数据库上进行查询的 SQL 执行截图

```
(base) LSY@MacBook-Air-51 database % psql tpch
psql (17.11 (Homebrew))
输入 "help" 来获取帮助信息.

tpch=# SELECT current_database();
 current_database
------------------
 tpch
(1 行记录)

tpch=# SELECT 'region' AS table_name, COUNT(*) AS row_count FROM region
UNION ALL SELECT 'nation', COUNT(*) FROM nation
UNION ALL SELECT 'supplier', COUNT(*) FROM supplier
UNION ALL SELECT 'customer', COUNT(*) FROM customer
UNION ALL SELECT 'part', COUNT(*) FROM part
UNION ALL SELECT 'partsupp', COUNT(*) FROM partsupp
UNION ALL SELECT 'orders', COUNT(*) FROM orders
UNION ALL SELECT 'lineitem', COUNT(*) FROM lineitem;
 table_name | row_count
------------+-----------
 region     |         5
 nation     |        25
 supplier   |       100
 customer   |      1500
 part       |      2000
 partsupp   |      8000
 orders     |     15000
 lineitem   |     60175
(8 行记录)

tpch=# SELECT r_regionkey, r_name
FROM region
ORDER BY r_regionkey;
 r_regionkey |          r_name
-------------+---------------------------
           0 | AFRICA
           1 | AMERICA
           2 | ASIA
           3 | EUROPE
           4 | MIDDLE EAST
(5 行记录)

tpch=# SELECT COUNT(*) AS order_count,
       ROUND(AVG(o_totalprice), 2) AS avg_totalprice
FROM orders;
 order_count | avg_totalprice
-------------+----------------
       15000 |      141826.46
(1 行记录)
```

```
tpch=# SELECT n.n_name, r.r_name
FROM nation AS n
JOIN region AS r
  ON n.n_regionkey = r.r_regionkey
ORDER BY n.n_nationkey
LIMIT 5;
          n_name           |          r_name
---------------------------+---------------------------
 ALGERIA                   | AFRICA
 ARGENTINA                 | AMERICA
 BRAZIL                    | AMERICA
 CANADA                    | AMERICA
 EGYPT                     | MIDDLE EAST
(5 行记录)

tpch=# SELECT o.o_orderkey,
       c.c_name,
       n.n_name AS nation,
       o.o_totalprice,
       o.o_orderdate
FROM orders AS o
JOIN customer AS c ON o.o_custkey = c.c_custkey
JOIN nation AS n ON c.c_nationkey = n.n_nationkey
ORDER BY o.o_orderkey
LIMIT 10;
 o_orderkey |       c_name       |          nation           | o_totalprice | o
------------+--------------------+---------------------------+--------------+--
          1 | Customer#000000370 | JAPAN                     |    172799.49 | 1
          2 | Customer#000000781 | CHINA                     |     38426.09 | 1
          3 | Customer#000001234 | ARGENTINA                 |    205654.30 | 1
          4 | Customer#000001369 | IRAN                      |     56000.91 | 1
          5 | Customer#000000445 | SAUDI ARABIA              |    105367.67 | 1
          6 | Customer#000000557 | MOROCCO                   |     45523.10 | 1
          7 | Customer#000000392 | PERU                      |    271885.66 | 1
         32 | Customer#000001301 | IRAN                      |    198665.57 | 1
         33 | Customer#000000670 | BRAZIL                    |    146567.24 | 1
         34 | Customer#000000611 | ROMANIA                   |     73315.48 | 1
(10 行记录)
```

实验总结

本实验完成了 MySQL、PostgreSQL 和 MariaDB 的安装与基本使用，并以 PostgreSQL 17.11 建立 TPC-H 数据库。通过 DDL 创建 8 张数据表并设置主外键约束，使用 \copy 按依赖顺序导入 SF=0.01 数据。经数据量统计、普通 SELECT 和 JOIN 查询验证，各表数据规模及关联结果均符合预期。通过实验进一步理解了数据库模式定义、完整性约束、数据导入和多表查询的基本流程。